LIÈGE université
Sciences Appliquées

Optimization and forecasting module overview

ELEN0445 - Microgrids

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You Have Learned

- What is a microgrid (structure)
- What are its main components
- Power electronics interfaces (MPPT, etc.)
- The levels of control
- How to design an installation using SMA's software

Optimal Control and Sizing

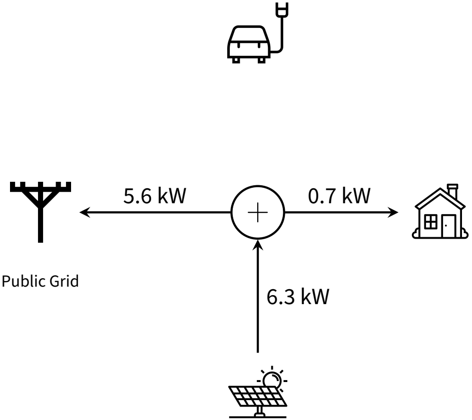
Objective: Learn how to optimally size and control a microgrid.

We will still focus on a one-node microgrid for illustration

Possible problem extensions:

- **Microgrid networks:** Adding power flow equations (spatial complexity).
- **Multi-energy systems:** District heating, Hydrogen (H_2), etc.
- **Community microgrids:** Fairness and sharing considerations.

Real-time Control:

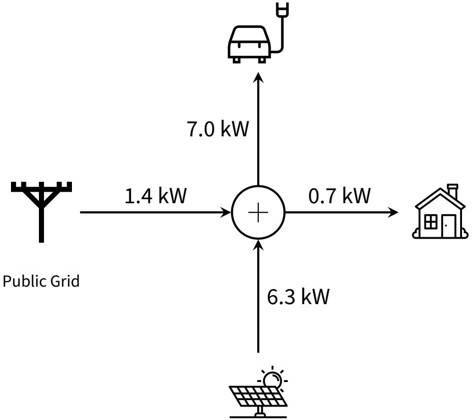


Max grid exchange: **11.0 kW**

Sequence of events:

1. Initial: Surplus exported to grid.

Real-time Control:

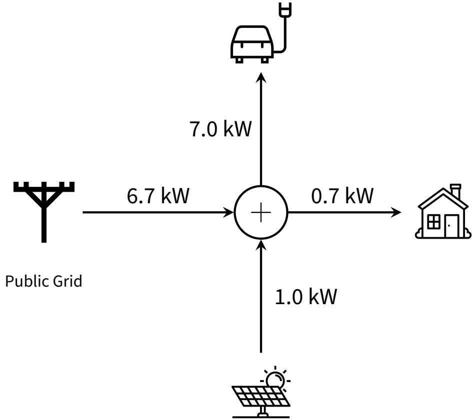


Max grid exchange: **11.0 kW**

Sequence of events:

1. Initial: Surplus exported to grid.
2. **Car connects**: Adding 7 kW load.

Real-time Control:

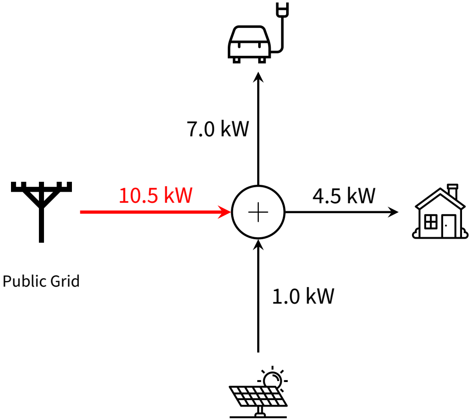


Max grid exchange: **11.0 kW**

Sequence of events:

1. Initial: Surplus exported to grid.
2. **Car connects**: Adding 7 kW load.
3. **Cloud passes**: PV drops to 1 kW.

Real-time Control:



Max grid exchange: **11.0 kW**

Sequence of events:

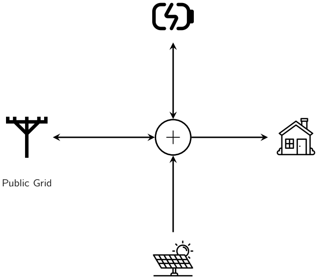
1. Initial: Surplus exported to grid.
2. **Car connects**: Adding 7 kW load.
3. **Cloud passes**: PV drops to 1 kW.
4. **Peak load**: House rises to 4.5 kW.

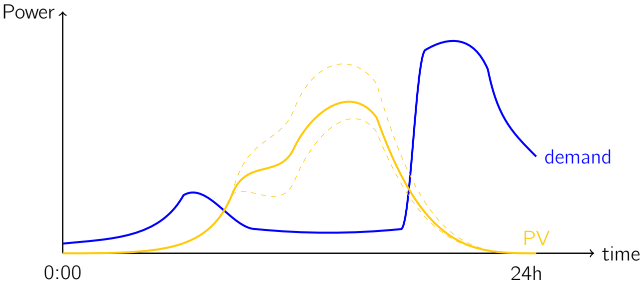
Decision Point

Import is at 10.5 kW. Any small increase will trip the 11 kW limit.

Should you stop charging?

Operational Planning (E.g. 24h ahead)





- Based on **forecasted demand and PV generation.**
- **Decisions:** Store, send back to grid, or move load?
- **Variables:** Function of prices, physical limits, and efficiencies.

Sizing (Investment Decisions)

Strategic investment questions:

- Should you connect to the grid? Which power?
- Should you buy a genset or a CHP (Combined Heat and Power)?
- Should you invest in PV or wind generation?
- Should you invest in storage?

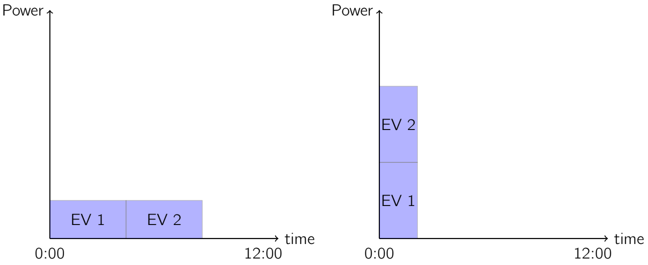
Interdependence

Sizing decisions depend directly on the operational planning policy.

Sizing depends on the operational planning policy

Example:

- 2 EVs and you need to choose how many Amps to take from the grid
- Can charge slow or fast
- Charge them simultaneously or one after the other.



Your ability to optimize charges will largely impact the grid connection.

- Good forecasts of prices, generation, and consumption required

Coordination of Time Scales

The global problem is decoupled in practice:

Horizon	**Decision Type**
Yearly	Investment (Sizing)
Hours/Days	Planning (Operational Planning)
Minutes/Seconds	Adjustment (Real-time Control)

Example: coordination between real-time control and operational planning

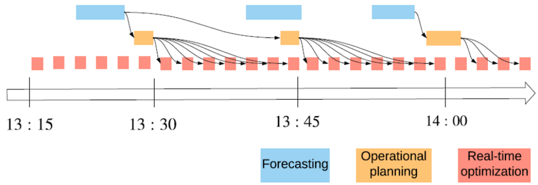


Fig. 1: Hierarchical control procedure illustration.

References:

- Dumas, J., Dakir, S., Liu, C., & Cornélusse, B. (2021). Coordination of operational planning and real-time optimization in microgrids. Electric Power Systems Research, 190, 106634.
- Moureau, C., Stegen, T., Glavic, M., & Cornélusse, B. (2025). A Predictive Flexibility Aggregation Method for Low Voltage Distribution System Control. ORBi-University of Liège.

Goals of this module

- Learn about mathematical programming
- Learn to model problems as LPs or MIPs
- Understand how solvers work
- Solve the real-time control problem
- Learn to make some forecasts
- Solve the operational planning problem
- Solve the sizing problem